MedInsure Health Claim Approval Model Presentation

3. Model Risk & Subpopulation Robustness

Sources of model risk in this scenario

- **Data risk**: Training data is not representative (rare surgeries underrepresented).
- **Model bias**: Systematically worse performance on certain claim categories.
- **Operational risk**: Missing features, runtime mismatches.
- **Security risk**: Manipulated diagnosis codes exploiting the model.
- **Reproducibility risk**: Missing logs and metadata.

3. Model Risk & Subpopulation Robustness

Why poor performance on subpopulations matters

- Real-world populations are diverse; ML must handle all groups reliably.
- Underrepresented categories leads to unpredictable failures.
- This can result in:
    1. **Incorrect approvals/denials**
    2. **Compliance issues**
    3. **Ethical and financial risks**

Hence, this case illustrates **the need for robustness testing across subpopulations** (e.g., claim type, hospital type, surgery category).

4. Stress Testing & API Stability

The case shows that under **5,000 claims/min**, the API slows and times out

Why stress testing is needed

- Production models face real-time, unpredictable loads.
- Stress testing reveals:
    1. **Latency bottlenecks**
    2. **Memory/CPU saturation**
    3. **Queue buildup**
    4. **Infrastructure fragility**

4. Stress Testing & API Stability

Methods to stabilize the API

- **Asynchronous processing with message queues**

  Use Kafka, RabbitMQ, or Celery for background claim scoring.

- **Horizontal scaling / autoscaling**

  Add more inference servers during peak times.

- ***(Optional additional methods)***

  Model distillation or optimization to reduce inference time

  Load balancing via API gateways

  Caching frequent predictions (if applicable)